Skills Summary

Composition Order and Inverse Domains

Use this reference while completing your worksheet or when studying.

Skill 1 — Order changes a composition

Read it inside out: $(f \circ g)(x) = f\big(g(x)\big)$ — the rule written *next to* the input acts first.

Not commutative: $f \circ g$ and $g \circ f$ are different rules. If a quoted answer looks wrong, compute both compositions; the wrong one is usually the other order.

Example: $f(x) = x + 4$, $g(x) = 3x$. Find $f\big(g(5)\big)$ and $g\big(f(5)\big)$.
Step 1. Each answer needs the inner rule evaluated first, so read each expression inside out and do that rule before the outer one.
Step 2. $g(5) = 15$, then $f(15) = 15 + 4 = 19$. As one rule, $f(g(x)) = 3x + 4$.
Step 3. $f(5) = 9$, then $g(9) = 27$. As one rule, $g(f(x)) = 3(x + 4)$.
$\Rightarrow \quad f\big(g(5)\big) = \mathbf{19}, \qquad g\big(f(5)\big) = \mathbf{27}$

Try it: $p(x) = x - 2$, $q(x) = x^2$. Find $p\big(q(3)\big)$.

check: 7

(!) Watch out: $(g \circ f)(x)$ with $f(x) = x - 4$ and $g(x) = x^2$ is $(x - 4)^2$, *not* $x^2 - 4$. Subtract first, square second.

Skill 2 — Testing a claimed inverse

The two-sided test: a rule is f^{-1} only when $f^{-1}\big(f(x)\big) = x$ *and* $f\big(f^{-1}(x)\big) = x$.

Fast refutation: pick one convenient input, run it through f, then through the claimed rule. If you do not get your input back, the claim is dead — one counterexample is enough.

Example: Test the claim that $\dfrac{x+3}{5}$ inverts $f(x) = 5x - 3$.
Step 1. A claimed inverse is worth testing with a single number before any algebra, so run one input out through f and back through the claim.
Step 2. $f(4) = 5(4) - 3 = 17$, then $\dfrac{17+3}{5} = \dfrac{20}{5} = 4$ — the input came back.
Step 3. In general $5\left(\dfrac{x+3}{5}\right) - 3 = (x + 3) - 3 = x$, so the other round trip works too.
$\Rightarrow$ round trip returns **4**, and the claim holds.

Try it: For $g(x) = 2x + 7$ and the claim $g^{-1}(x) = \dfrac{x-7}{2}$, run the input 6 out and back.

check: 6

Skill 3 — The domain an inverse inherits

Inputs of f^{-1} are outputs of f. Solving for y gives the formula; the *range* of the original gives the restriction the formula must carry.

Square roots: $\sqrt{\ }$ is never negative, so a rule built from $\sqrt{x-h}+k$ has outputs $x \geq k$ — and its inverse only accepts inputs $x \geq k$.

Example: Find f^{-1} for $f(x) = \sqrt{x-1}+2$, and state its domain.
Step 1. The formula comes from solving $x = f(y)$ for y, and the restriction comes separately from asking what outputs f can actually produce.
Step 2. $x - 2 = \sqrt{y-1} \;\Rightarrow\; (x-2)^2 = y - 1$
Step 3. Outputs of f: the root is at least 0, so $f(x) \geq 2$. The inverse therefore accepts only $x \geq 2$.

$\Rightarrow \quad \boldsymbol{f^{-1}(x) = (x-2)^2 + 1}$ for $x \geq 2$

Try it: Find g^{-1} for $g(x) = \sqrt{x+4} - 1$ (and say what its inputs must satisfy).

check: $\boldsymbol{g^{-1}(x) = (x+1)^2 - 4}$, inputs $x \geq -1$